The Impact of Strict Voter Photo ID Laws on Turnout

MA592 - Causal Inference Project

Alen Abdildayev

1. Introduction

In order to vote in elections in the United States, some states have implemented strict voter photo ID laws, where voters are required to have a photo ID to participate. Voter ID requirements are quite popular among Americans; proponents argue that they increase voter confidence and decrease electoral fraud. However, opponents of these laws argue that there is a lack of evidence of meaningful fraud and that these laws are targeted at specific groups of people.

In this project, we ask specifically whether the implementation of strict photo ID laws impacts turnout in elections. In order to evaluate this, we will estimate the causal effect of these laws by comparing the change in voter participation in states that adopted them against baseline turnout trends, by using a Staggered Difference-in-Differences framework.

Because of the complexity of this idea, we cannot rule out unobserved confounders and time-varying confounders. Nonetheless, it is still highly valuable to evaluate the impact of these laws in the United States. This question becomes more interesting when we shift our focus from the overall picture to specific demographic groups, even if aggregate turnout appears unchanged.

2. Data Set

The data set was taken from the Current Population Survey (CPS) Voter Supplement. The CPS is a monthly survey with a Voter Supplement administered in November of election years, which is why we used only November of even years in our data. The unit of analysis is the eligible voter: individuals who are 18 years or older and hold U.S. citizenship. The problem is that the CPS relies on self-reported data, so we are assuming that individuals' responses are accurate.

Our treatment is the implementation of a strict voter photo ID law. Note that there are states with non-strict voter photo ID laws, so to be precise, it is more about "strictness." Our outcome is turnout, representing whether the individual respondent voted in that year's election.

In order to isolate the causal effects of the laws, we introduce the following main covariates of individual and geographic factors: SEX: Categorical variable for voter identified sex; RACE/HISPAN: Categorical variable for racial/ethnic identity; EDUC: Categorical variable for educational attainment; FAMINC: Categorical variable for family income bracket; AGE: Categorical variable for age cohort; CITIZEN: Categorical variable for citizenship status; LABFORCE: Categorical variable for labor force participation; METRO: Categorical variable for urbanicity/metropolitan status; STATEFIP: State fixed effects, proxy for time-invariant state characteristics; YEAR: Year fixed effects, proxy for national temporal shocks;

A potential issue is that some states (such as Texas) implemented this law in 2013, while we count only election years (even years). Thus, we are "shifting" odd years to even years (e.g., we will assume Texas implemented the law in 2014).

3. Methods and Analysis

3.1 Staggered Difference-In-Differences

In our research we will use Staggered Difference-In-Differences framework. The reasoning behind that is that each state passes this law at different years, so using normal Difference-In-Differences is not enough. Our parameter of interest would be Average Treatment Effect on the treated (ATT)

$$Y_{(i,s,t)} = \alpha + \beta D_{(s,t)} + X'_{(i,s,t)}\gamma + \mu_s + \lambda_t + \epsilon_{(i,s,t)}$$

where: $Y_{(i,s,t)}$: our outcome; β: ATT; $D_{(s,t)}$: treatment indicator; $X_{(i,s,t)}$: vector of covariates; μ_s: state fixed effects; λ_t: year fixed effects; $\epsilon_{(i,s,t)}$: error term.

Our Assumption are: Parallel Trends: In the absence of the law, the turnout in treated states would have same trajectory as the turnout in control states; No Anticipation Effect: Voters do not change their behavior in the years prior to law's implementation; SUTVA: Implementation of a strict ID law in one state must not affect the voting behavior of other states;

3.2 Subgroups

However, since we also want to check effect of laws on turnout in subgroups, it means we have another parameter of interest: Conditional Average Treatment Effects on the Tread (CATT)

$$Y_{(i,s,t)} = \alpha + \sum_{k=1}^{K} \beta_k (D_{(s,t)} \times G^k_{(i,s,t)}) + \sum_{k=1}^{K} \delta_k G^k_{(i,s,t)} + X'_{(i,s,t)}\gamma + \mu_s + \lambda_t + \epsilon_{(i,s,t)}$$

where: $G^k_{(i,s,t)}$: specific demographic category, β_k: group-specific ATT, δ_k: group-specific baseline fixed effect

3.3 Weighted Cluster-Robust Variance Estimator

Since individuals in our data are grouped by state, it means they are not entirely independent. Because of that we cluster standard errors at the state level to account for the non-independence of voters, preventing the severe downward bias in standard errors.

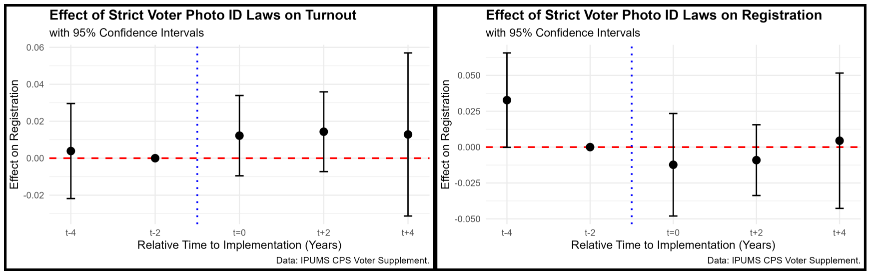
$$\hat{V}_{CRVE}(\hat{\beta}) = c\,(X'WX)^{-1}\left(\sum_{g=1}^{G} X'_g \hat{u}_g \hat{u}'_g X_g\right)(X'WX)^{-1}$$

where c: correction factor - $\frac{G}{G-1} \times \frac{N-1}{N-K}$, W: diagonal matrix of the survey weights, $\hat{u}_g$: Vector of weighted residuals.

4. Results

4.1 Overall Results

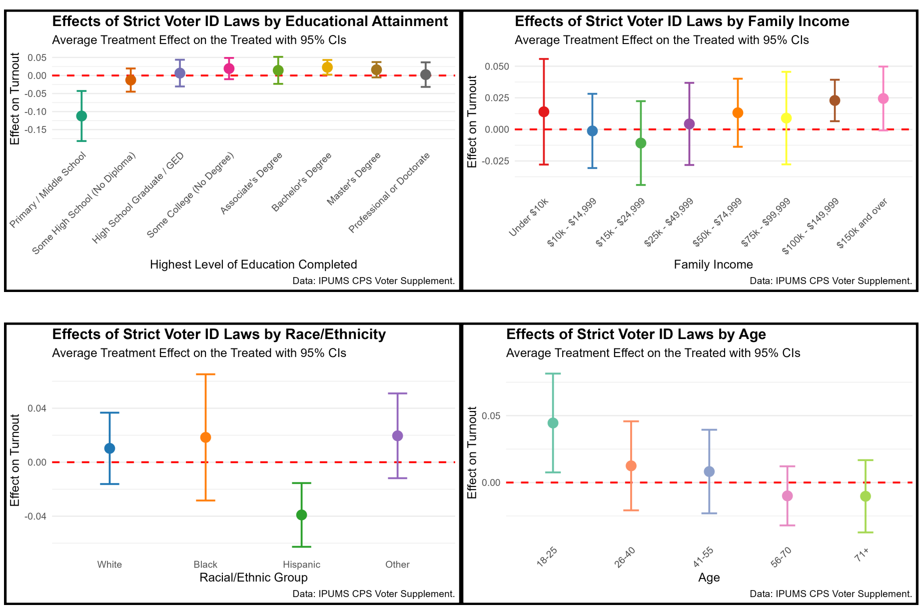
Our baseline treatment effect has a value of 0.00981 with a standard error of 0.01440, which indicates that these laws do not significantly impact overall turnout. Even when examining specific relative years, the results remain consistent. Additionally we would like to know how Strict Photo ID laws impact on registration to vote.



One positive finding is that for overall turnout, our $t-4$ value does not significantly deviate from zero, which means our parallel trends assumption is not violated. However for registration case, the estimate for $t-4$ is significant enough to not be zero, which means it violates parallel trends assumption, which means we can't use Staggered Difference-in-Differences for it.

4.2 Subgroup Results

However, even though there is no statistical significant difference in overall turnout, it might be not the case for specific subgroups. We will look how laws affect each of the next eight groups: Race, Family Income, Education, Age, Sex, Citizenship, Urbanicity and Labor force.



Education	Effect	Lower.CI	Upper.CI
Primary/Middle	−0.11231	−0.18171	−0.04292
Some−HS	−0.01267	−0.04484	0.01950
High School	0.00665	−0.03037	0.04367
College	0.01923	−0.01021	0.04867
Associate	0.01423	−0.02321	0.05168
Bachelor's	0.02268	0.00234	0.04303
Master's	0.01599	−0.00525	0.03723
PhD	0.00244	−0.03165	0.03652

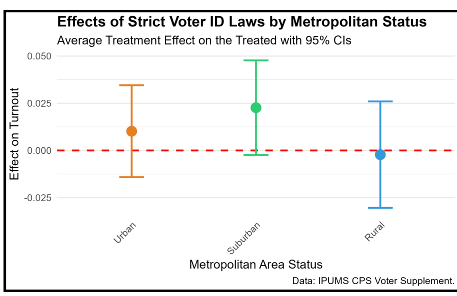
Income	Effect	Lower.CI	Upper.CI
<$10k	0.01395	−0.02785	0.05575
10−15k	−0.00122	−0.03060	0.02815
15−25k	−0.01082	−0.04399	0.02234
25−50k	0.00433	−0.02817	0.03682
50−75k	0.01316	−0.01378	0.04011
75−100k	0.00893	−0.02768	0.04555
100−150k	0.02288	0.00648	0.03928
150k+	0.02446	−0.00080	0.04973

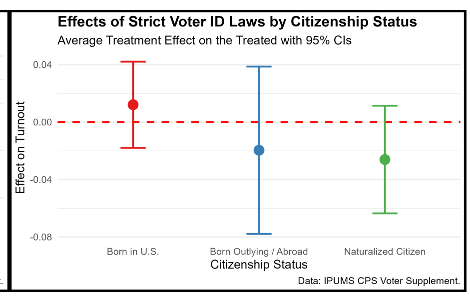
Race	Effect	Lower.CI	Upper.CI
White	0.01026	−0.01617	0.03670
Black	0.01840	−0.02838	0.06517
Hispanic	−0.03907	−0.06270	−0.01545
Other	0.01958	−0.01184	0.05099

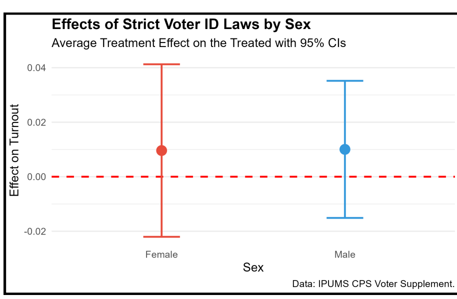
Age	Effect	Lower.CI	Upper.CI
18−25	0.04451	0.00757	0.08144
26−40	0.01248	−0.02080	0.04576
41−55	0.00820	−0.02304	0.03944
56−70	−0.00998	−0.03208	0.01212
71+	−0.01029	−0.03731	0.01673

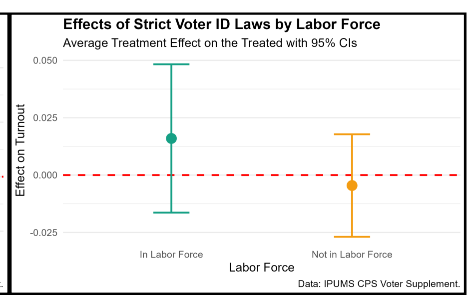
From the tables above, we can see that strict photo ID laws across racial groups impact only Hispanic voters negatively, which makes sense. The same applies to young voters, it is plausible that they would have higher turnout following the implementation of these laws, as they may believe the system is more secure. Additionally, it makes sense that individuals with lower levels of education would have lower turnout, as they may face difficulties obtaining photo IDs in the first place.

However, what I find curious is the positive turnout for high-income families; specifically those with an annual salary of $100k+, and for voters who have a Bachelor's degree, but not a Master's or PhD.









Urbanicity	Effect	Lower.CI	Upper.CI
Urban	0.01017	−0.01417	0.03452
Suburban	0.02265	−0.00241	0.04771
Rural	−0.00222	−0.03041	0.02596

Citizenship	Effect	Lower.CI	Upper.CI
U.S. Born	0.01215	−0.01783	0.04212
Abroad	−0.01958	−0.07785	0.03868
Naturalized	−0.02604	−0.06353	0.01145

Sex	Effect	Lower.CI	Upper.CI
Male	0.01003	−0.01512	0.03518
Female	0.00960	−0.02205	0.04126

Labor.Force	Effect	Lower.CI	Upper.CI
In Force	0.01596	−0.01637	0.04830
Not in Force	−0.00456	−0.02691	0.01779

When examining other subgroups, such as sex, citizenship, urbanicity, and labor force status, there are no significant deviations in turnout from zero. However, it is worth noting that suburban turnout is nearly significant, but it still remains statistically insignificant because its confidence interval includes zero.

4.3 Sensitivity Analysis

Now, we would like to know the sensitivity of our analysis. We would do this by removing one of the states and checking effect on overall turnout again.

State	Effect	Lower.CI	Upper.CI
Georgia	−0.00326	−0.02133	0.01480
Indiana	0.01226	−0.01919	0.04370
Kansas	0.01002	−0.01961	0.03965
Tennessee	0.01571	−0.01310	0.04453
Mississippi	0.01102	−0.01856	0.04059
Texas	0.01268	−0.01953	0.04489
Virginia	0.00632	−0.02421	0.03685
Wisconsin	0.01351	−0.01651	0.04354
Arkansas	0.01199	−0.01695	0.04092
North Carolina	0.00909	−0.02051	0.03870
Ohio	0.00930	−0.02063	0.03924

As we can see from the results above, the strict Photo ID laws still do not significantly impact overall turnout, even if we remove each of the state.

5. Conclusion

In conclusion, there is no evidence that strict Photo ID laws statistically significantly impact election turnout. However, analyzing specific demographic groups reveals notable shifts. For instance, following the implementation of these laws, Hispanic and less-educated voters participated at lower rates, whereas high-income families, individuals with bachelor's degrees and younger voters demonstrated higher turnout. In future research, incorporating additional covariates, such as individual political affiliation would provide more nuanced insights. Additionally, expanding the scope to include other legislative frameworks, such as non-strict voter ID laws, would further clarify the impact of these policies.